Generator Architecture: Block-by-Block Specification

$$G_\theta : \mathbb{R}^{128} \longrightarrow [-1, 1]^{1\times 64\times 64}$$

Block-by-block reading

The diagram is read top-to-bottom. *Green* blocks denote input/output tensors that live at the boundary of the network and that an external consumer is meant to observe (z and $G_\theta(z)$). *Blue* blocks denote internal hidden activations $\mathbf{h}^{(\ell)}$ that the network materializes during the forward pass and discards afterward. *Amber* blocks denote the parameterized operators that map one tensor state to the next. Side labels on the left record the spatial resolution $H^{(\ell)} \times W^{(\ell)}$ at each layer, making the doubling pattern $4 \to 8 \to 16 \to 32 \to 64$ explicit. Below we walk through every block in the order in which the forward pass visits them.

Block 1 — Latent input $z \in \mathbb{R}^{128\times 1\times 1}$

The latent vector z is drawn from an isotropic Gaussian $\mathcal{N}(\mathbf{0}, \mathbf{I}_{128})$ with $d_z = 128$. It is presented to the network not as a flat 128-vector but as a degenerate 3-D tensor with shape $128 \times 1 \times 1$, i.e. 128 channels at a 1×1 spatial grid. This reshape is a notational convenience that lets the first stage be written as a transposed convolution rather than a linear projection followed by a reshape; the two are arithmetically equivalent when the kernel and output spatial size both equal four and stride is one. The choice of an isotropic Gaussian prior is standard for W-GAN-GP: the Kantorovich–Rubinstein duality used to derive the gradient-penalty objective requires only that the prior have a tractable density and full support on its support set, and $\mathcal{N}(\mathbf{0}, \mathbf{I}_{128})$ satisfies both.

Block 2 — Stage 1 operator $\mathrm{ConvT}^{128\to 512}_{k=4,\, s=1,\, p=0} \circ \mathrm{BN} \circ \sigma_{\mathrm{ReLU}}$

Stage 1 is the only stage that does *not* double the spatial resolution; its job is to lift the latent vector into a small but high-channel spatial grid. The transposed convolution $W_1 \in \mathbb{R}^{128\times 512\times 4\times 4}$ with stride 1 and zero padding expands the 1×1 spatial extent of z to 4×4 while contracting the channel count from a high latent dimensionality (128) to an even higher feature-channel count (512). Intuitively, the network is buying *spatial expressiveness* at the cost of *channel expressiveness*; subsequent stages reverse this trade. The output is then passed through batch normalization $\mathrm{BN}_1(\cdot; \gamma_1, \beta_1)$, which centres and scales each channel across the mini-batch, and a pointwise ReLU activation $\sigma_{\mathrm{ReLU}}(x) = \max(0, x)$.

Block 3 — Hidden state $\mathbf{h}^{(1)} \in \mathbb{R}^{512\times 4\times 4}$

This is the first internal feature map the generator produces. It contains $512 \cdot 4 \cdot 4 = 8\,192$ scalars, organised as 512 spatial "feature pictures" on a 4×4 grid. The high channel count gives the network enough capacity to encode, in distributed form, the relevant modes of the target distribution; the small spatial extent makes that encoding compact. In a recurrence-plot context, this layer can be thought of as the network's coarsest decision about *global recurrence structure* — whether the plot is diagonal-dominated, block-dominated, or speckle-dominated — before any fine-grained spatial detail is committed.

Block 4 — Stage 2 operator $\mathrm{ConvT}^{512\to 256}_{k=4,\, s=2,\, p=1} \circ \mathrm{BN} \circ \sigma_{\mathrm{ReLU}}$

From Stage 2 onward, the architectural pattern stabilises: stride 2, padding 1, kernel 4, channel halving. This is the canonical DCGAN upsampling block. The stride-2 transposed convolution doubles each spatial dimension; with output padding implicit in the $(k, s, p) = (4, 2, 1)$ choice, the input 4×4 becomes 8×8 *exactly*, with no off-by-one artefacts. The channel halving $512 \to 256$ is a budgeted re-allocation of capacity from channels to space: as the feature maps grow, each individual feature plane carries less channel-specific information but a richer spatial pattern. The BN and ReLU follow for the same reasons as in Stage 1.

Block 5 — Hidden state $\mathbf{h}^{(2)} \in \mathbb{R}^{256\times8\times8}$

This is the second internal feature map, with $256 \cdot 8 \cdot 8 = 16\,384$ scalars. The shape doubling is the first instance of the *progressive-upsampling invariant* that holds throughout the rest of the network: each subsequent stage takes the previous spatial extent and doubles it while halving the channel count, keeping the total tensor volume roughly constant in this layer (16 384 vs. 8 192 in $\mathbf{h}^{(1)}$, growing by exactly a factor of two; the same factor will recur). At this resolution the network begins to commit to medium-scale recurrence-plot structures such as diagonal bands and rectangular regimes.

Block 6 — Stage 3 operator $\mathrm{ConvT}^{256\to128}_{k=4,\,s=2,\,p=1} \circ \mathrm{BN} \circ \sigma_{\mathrm{ReLU}}$

Structurally identical to Stage 2 except for the channel widths. The transposed convolution again doubles spatial extent ($8 \to 16$) and halves channels ($256 \to 128$). The choice to keep the kernel size fixed at 4 across all upsampling stages is a deliberate design choice: a stride-2, kernel-4, padding-1 transposed convolution has a receptive field of two input cells per output cell, which gives every output position information from two distinct input positions and avoids the checkerboard artefacts that a stride-2-kernel-3 configuration is known to produce.

Block 7 — Hidden state $\mathbf{h}^{(3)} \in \mathbb{R}^{128\times16\times16}$

Third internal feature map, $128 \cdot 16 \cdot 16 = 32\,768$ scalars. By this layer the network has committed to a 16×16 approximation of the recurrence plot, which in 64-pixel terms corresponds to a coarse-quantised version of the target image. Empirically, mode-coverage diagnostics computed against generators at this depth tend to detect diversity correctly but cannot distinguish between near-duplicates at the finest scale.

Block 8 — Stage 4 operator $\mathrm{ConvT}^{128\to64}_{k=4,\,s=2,\,p=1} \circ \mathrm{BN} \circ \sigma_{\mathrm{ReLU}}$

The penultimate upsampling stage. Channel halving $128 \to 64$ and spatial doubling $16 \to 32$ continue the canonical pattern. This is the last stage that uses batch normalisation; the terminal stage drops BN deliberately so that the final activation can centre at any value the target distribution prefers without being forced through a per-channel zero-mean constraint. The BN here is still active because the layer is still operating in feature space, not output space.

Block 9 — Hidden state $\mathbf{h}^{(4)} \in \mathbb{R}^{64\times32\times32}$

The fourth and final internal feature map, $64 \cdot 32 \cdot 32 = 65\,536$ scalars. The activations at this layer have the same spatial extent as a half-resolution recurrence plot and 64 channels of feature information. Empirically, this is where most of the high-frequency texture of the target recurrence plot is encoded; the final stage's role is mainly to project this rich feature representation down to one channel and apply the squashing nonlinearity.

Block 10 — Stage 5 operator $\mathrm{ConvT}^{64\to1}_{k=4,\,s=2,\,p=1} \circ \tanh(\cdot)$

The terminal stage differs from the previous four in two respects: (i) it collapses the 64-channel feature map to a single output channel, matching the grayscale nature of the target recurrence plot; (ii) it replaces $\mathrm{BN} \circ \sigma_{\mathrm{ReLU}}$ with a pointwise $\tanh(\cdot)$, which bounds the per-pixel output to the open interval $(-1, 1)$. The tanh choice is consistent with the per-bucket min–max normalisation applied to the training tensors, which scales every training recurrence plot to $[-1, 1]$ via an invertible $(\min, \max)$-tuple. Because the generator's output is bounded by the same interval as the training data, the per-pixel histograms of $G_\theta(z)$ and a training tensor live on the same support, which is a precondition for the Wasserstein-1 distance estimator to be well-defined under the W-GAN-GP duality. Note also that there is no batch-normalisation at this stage: applying BN immediately before tanh would impose a per-channel zero-mean constraint on the output, biasing the network away from high-amplitude plots even when the target distribution favours them.

Block 11 — Generated sample $G_\theta(z) \in [-1, 1]^{1\times64\times64}$

The output is a single-channel image of size 64×64 with per-pixel values in $[-1, 1]$. Multiplying by the persisted $(\min, \max)$ tuple of any training bucket inverts the per-bucket normalisation and recovers a distance-plot value with the same units as the original Euclidean pairwise distance. The output passes downstream into three independent post-hoc analyses without further parameterisation: (i) the critic D_ϕ, which scores it during training; (ii) the mode-coverage nearest-neighbour search, which compares it against the training tensor bank at each 5,000-step checkpoint; and (iii) optional visual inspection or downstream regime classification, which can use $G_\theta(z)$ as a synthetic recurrence-plot data point.

Side annotations and arrows

The five labels on the left edge of the pipeline (4×4, 8×8, 16×16, 32×32, 64×64) record the spatial resolution of the hidden state immediately to their right. They make visible at a glance the geometric-progression structure $H^{(\ell)} = 2H^{(\ell-1)}$ for $\ell = 2, \ldots, 5$ that distinguishes Stages 2–5 from the dimensional-lifting Stage 1. The arrows between blocks denote ordinary forward-pass data flow: at training time the adjoint flow (backpropagation) traverses the same edges in reverse, collecting gradients with respect to every parameter in $\theta = \bigcup_{\ell=1}^{5} \{W_\ell, b_\ell\} \cup \bigcup_{\ell=1}^{4} \{\gamma_\ell, \beta_\ell\}$.

Parameter accounting

A complete count of the learnable parameters in G_θ illustrates the budget choices implicit in the architecture. The five transposed convolutions contribute, by stage, $128 \cdot 512 \cdot 4 \cdot 4 = 1\,048\,576$, $512 \cdot 256 \cdot 4 \cdot 4 = 2\,097\,152$, $256 \cdot 128 \cdot 4 \cdot 4 = 524\,288$, $128 \cdot 64 \cdot 4 \cdot 4 = 131\,072$, and $64 \cdot 1 \cdot 4 \cdot 4 = 1\,024$ weight scalars respectively (plus a small bias per output channel that we omit). The four BN layers contribute $2(c_1 + c_2 + c_3 + c_4) = 2(512 + 256 + 128 + 64) = 1\,920$ scalars. Roughly 3.8 million learnable parameters in total. Stage 2 dominates the budget because both its input and output channel counts are large; stage 5 is essentially free. This concentration of capacity in the middle layers is intentional: the early-middle stages of a transposed-convolutional generator are where the bulk of distributional structure is committed, while the late stages mainly perform geometric refinement.

Why this exact configuration

The architecture is a faithful instantiation of the DCGAN/W-GAN-GP generator family, with three deliberate choices recorded here so that they can be revisited as ablations rather than absorbed silently into the spec. **Latent dimension** 128**:** a standard choice for 64×64 image generation, large enough to provide capacity for the target distribution without inflating the optimisation landscape's condition number. **Kernel size** 4 **with stride** 2 **and padding** 1**:** chosen over $(3, 2, 1)$ to avoid checkerboard artefacts and over $(5, 2, 2)$ to keep parameter count manageable; the $(4, 2, 1)$ choice is the one for which every output position sees an evenly tiled receptive field of two input cells. **No batch-normalisation in the terminal stage:** explicitly motivated so that the final tanh can produce arbitrary-mean outputs aligned with the per-bucket min–max-normalised training distribution. None of these three choices is novel; the contribution is the locked, pre-registered combination that allows the seed-variance of training to be measured on a stable architectural baseline.

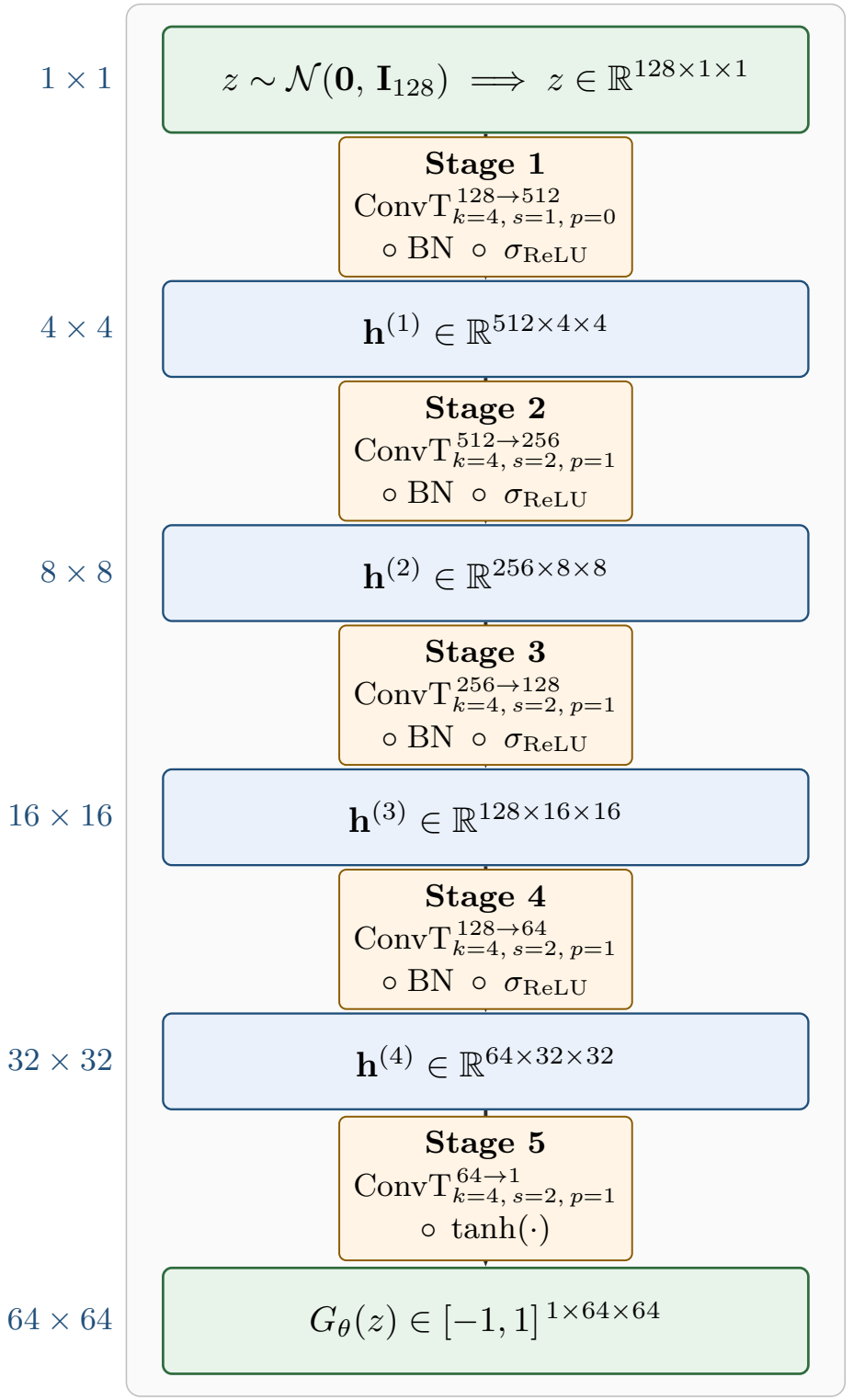


Figure 1: Generator G_θ as a deterministic deep transposed-convolutional map from latent code z to a single-channel 64×64 image.